Developing Open, Interoperable Semantic Resources

Tony Awards Winners Ontology

GRUPO 10

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1. Introduction

This report describes the development of the Tony Awards Winners

Ontology, an OWL ontology designed to represent information about Tony Award winning records. The model covers winners, awarded works, award categories, gender categories and award years, using data from Tony Awards winners between 1982 and 2015.

The final project includes an OWL ontology, SHACL validation shapes, RDF data, ontology requirements, evaluation evidence and this report. The report follows the structure requested in the final assignment: introduction, development, requirements, reuse, evaluation, documentation, FAIRness, mapping with another group ontology and conclusions.

2. Development methodology

The work follows the main phases of the LOT methodology: ontology requirements specification, ontology implementation, ontology evaluation and ontology publication. First, functional requirements were identified from the source data and use cases. Then, a conceptual model was designed and implemented in OWL. After that, SHACL shapes were created to validate the RDF data, and the ontology was evaluated with OOPS! and FOOPS!.

The work was iterative: initial conceptualization and evaluation results were used to improve the ontology, especially by introducing the central class myOnt:AwardWinning as an n-ary relation, adding inverse properties, improving metadata and correcting validation data

3. Ontology requirements

The ontology requirements were derived from the original Tony Awards dataset fields: award category, year, winner name, awarded work and gender category. These requirements guide the ontology scope and justify the central award-winning record model.

1	Identifier	Requirement	Provenance	Comments
2	FR01	List all Tony Award winners for a given year.	YEAR, NAME, AWARD_CATEGORY	Basic query by year.
3	FR02	Retrieve the winner of a specific award category in a given year.	AWARD_CATEGORY, YEAR, NAME	Category-year lookup.
4	FR03	List all award categories present in the dataset.	AWARD_CATEGORY	Identify available award types.
5	FR04	Retrieve all works for which a specific person won a Tony Award.	NAME, WORK	Person-work relationship.
6	FR05	Count how many Tony Awards a person has won.	NAME, YEAR	Aggregate per person.
7	FR06	Identify people who have won the award multiple times.	NAME, YEAR	Detect repeated winners.
8	FR07	List all works that have received a Tony Award.	WORK, AWARD_CATEGORY	Production-based analysis.
9	FR08	Retrieve the winner associated with a specific work.	WORK, NAME	Reverse lookup from work to person.
10	FR09	Retrieve the winners for a specific award category across different years.	AWARD_CATEGORY, YEAR, NAME	Historical analysis of categories.
11	FR10	Determine the earliest and latest years represented in the dataset.	YEAR	Dataset temporal coverage.
12	FR11	List all winners belonging to the feminine category.	GENDER_CATEGORY, NAME	Gender filtering.
13	FR12	List all winners belonging to the masculine category.	GENDER_CATEGORY, NAME	Gender filtering.
14	FR13	Count the number of awards won by each gender category.	GENDER_CATEGORY, NAME	Gender distribution analysis.
15	FR14	Retrieve all winners of a given work across different award categories.	WORK, AWARD_CATEGORY, NAME	Work-centered exploration.
16	FR15	Identify works that have won awards in multiple years.	WORK, YEAR	Detect recurring productions.
17	FR16	Retrieve all winners within a specific award type, Play or Musical.	AWARD_CATEGORY, NAME	Distinguish theatre vs musical awards.
18	FR17	List winners by gender category for a specific year.	YEAR, GENDER_CATEGORY, NAME	Gender comparison per year.
19	FR18	Generate statistics of awards per year.	YEAR, AWARD_CATEGORY	Annual aggregation.
20	FR19	Detect duplicate records with the same year, award category and winner.	YEAR, AWARD_CATEGORY, NAME	Data quality validation.
21	FR20	Retrieve the complete historical record of a given award category ordered by year.	AWARD_CATEGORY, YEAR, NAME, WORK	Full historical timeline of the category.

The requirements cover basic retrieval tasks, historical analysis, gender-category filtering, work-centered exploration, duplicate detection and summary statistics.

4. Data analysis and non-ontological resources

The main non-ontological resources are CSV datasets about Tony Award winners. The repository includes an initial actress dataset, an actor dataset and a combined dataset. The combined dataset uses the fields PREMIO, AÑO, NOMBRE, OBRA and CATEGORIA, which were mapped to award category, year, winner, work and gender category.

No core data field from the combined model was discarded for the ontology design. Blank or incomplete rows are not suitable for conformant RDF validation because the SHACL shapes require a winner, work, award category, year and gender category for each award-winning record.

Resource	Purpose
data/dataset inicial y dataset	Initial dataset for actress award

adicional/dataset inicial actrices.csv	winners
data/dataset inicial y dataset adicional/dataset actores.csv	Additional dataset for actor award winners
data/dataset_combinado_.csv	Combined source dataset with award category, year, winner, work and gender category.
data/tony_awards_data.ttl	RDF data generated according to the ontology.

5. Reused ontological resources

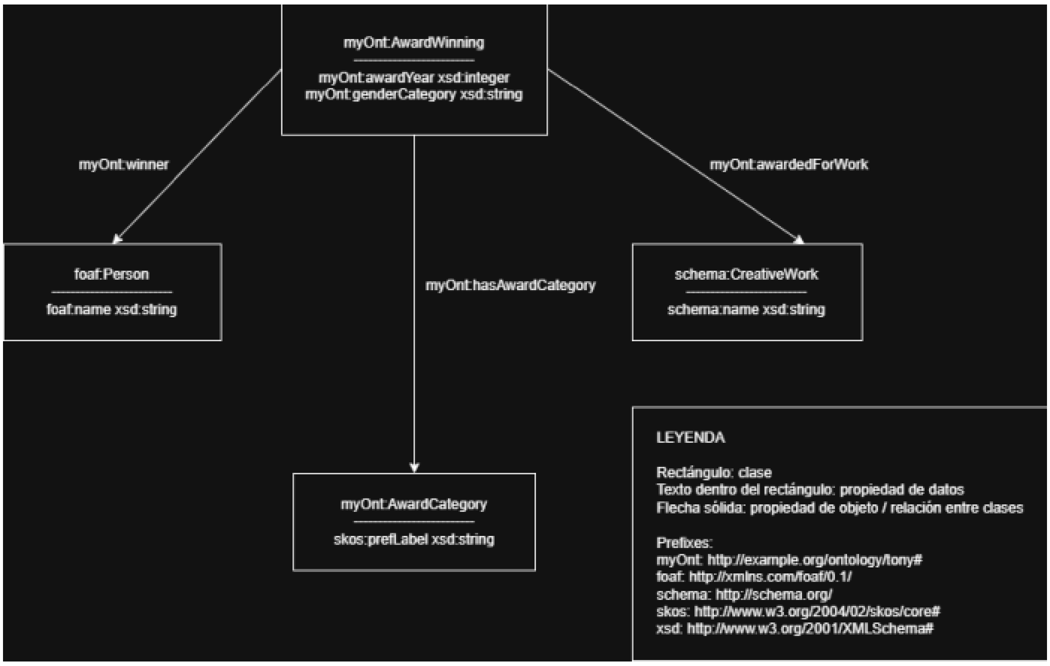
In ontology/Ontology.ttl and ontology/Ontology.owl, the ontology reuses well-known vocabularies for people, creative works, labels and metadata. No complete Tony Awards domain ontology was found in the project repository, so the domain model was created for this project while reusing standard vocabularies when appropriate.

Resource	Terms reused	Purpose
FOAF	foaf:Person, foaf:name	Representation of award winners as people and their names
Schema.org	schema:CreativeWork	Representation of plays and musicals as creative works
SKOS	skos:prefLabel	Preferred labels for award and gender category resources.
Dublin Core Terms	dcterms:title, dcterms:creator, dcterms:license, etc.	Ontology metadata and provenance.
VANN	vann:preferredNamespacePrefix, vann:preferredNamespaceUri	Namespace metadata.

6. Conceptual model and design decisions

The main modelling decision is to represent each award-winning entry as an n-ary relation. A direct relation between a person and an award is not enough, because each record also includes year, work, award category and gender category. For this reason, the central class is myOnt:AwardWinning.

Each instance of myOnt:AwardWinning links one winner, one creative work, one award category, one gender category and one award year. This addresses the feedback received in the conceptualization phase about clarifying which actor receives a prize and whether the prize is associated with a work.



7. Ontology implementation

The ontology was implemented in OWL and serialized in Turtle and RDF/XML. The repository contains ontology/Ontology.ttl and ontology/Ontology.owl. The implementation declares 14 classes, 10 object properties and 7 datatype properties.

Item	Value
Ontology URI	https://w3id.org/tony-awards/ontology (W3ID PR pending)
Namespace	https://w3id.org/tony-awards/ontology#
Version IRI	https://w3id.org/tony-awards/ontology/1.0.0
Turtle serialization	ontology/Ontology.ttl
RDF/XML serialization	ontology/Ontology.owl

Declared license	Creative Commons Attribution 4.0

A example of metadata:

```
<https://w3id.org/tony-awards/ontology>
    a owl:Ontology ;

    dcterms:title "Tony Awards Winners Ontology"@en ;
    dcterms:creator "Group 10"@en ;
    dcterms:license <https://creativecommons.org/licenses/by/4.0/> ;
    owl:versionIRI <https://w3id.org/tony-awards/ontology/1.0.0> ;
    owl:versionInfo "1.0.0" .
```

8. Knowledge Graph

The RDF data is stored in data/tony_awards_data.ttl. In the repository version reviewed for this report, this file is a conformant RDF sample containing two award-winning records, two people, two creative works, two award categories and two gender categories. It is aligned with the ontology and with the SHACL shapes.

The combined CSV contains more rows than the current RDF sample. A full RDF materialization of all complete CSV rows would be a useful improvement if the project is extended.

```
ex:Winning_2015_Helen_Mirren
  a myOnt:AwardWinning ;
  myOnt:recordIdentifier "Winning_2015_Helen_Mirren" ;
  myOnt:winner ex:Helen_Mirren ;
  myOnt:awardedForWork ex:The_Audience ;
  myOnt:hasAwardCategory ex:Best_Actress_Play ;
  myOnt:hasGenderCategory ex:Feminine ;
  myOnt:awardYear 2015 ;
  myOnt:genderCategory "femenina" .
```

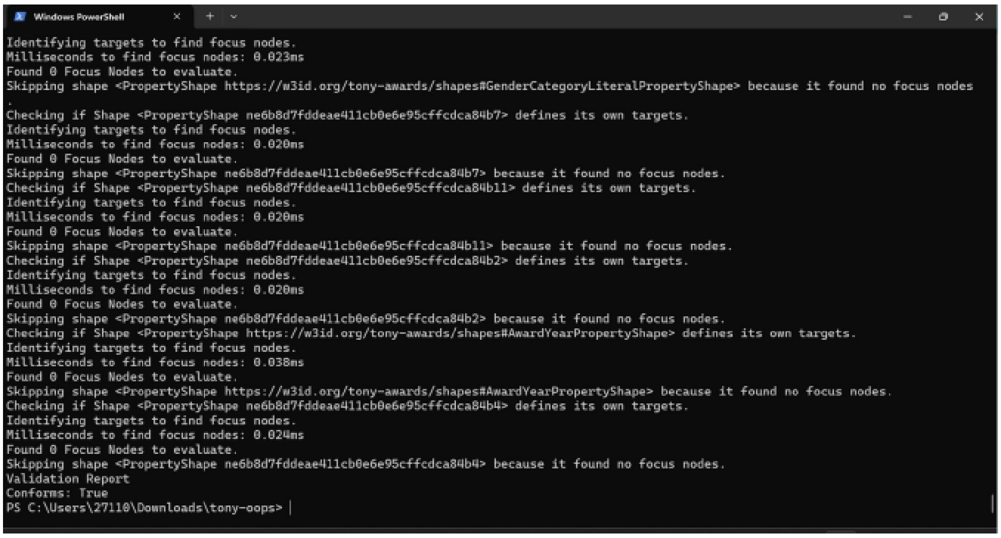
9. SHACL shapes and validation

The SHACL shapes are stored in shacl/shapes.ttl. They validate cardinality, datatype, class membership, ranges and controlled values. The validation report is stored in shacl/report.ttl and states that the data conforms.

Command used for validation:

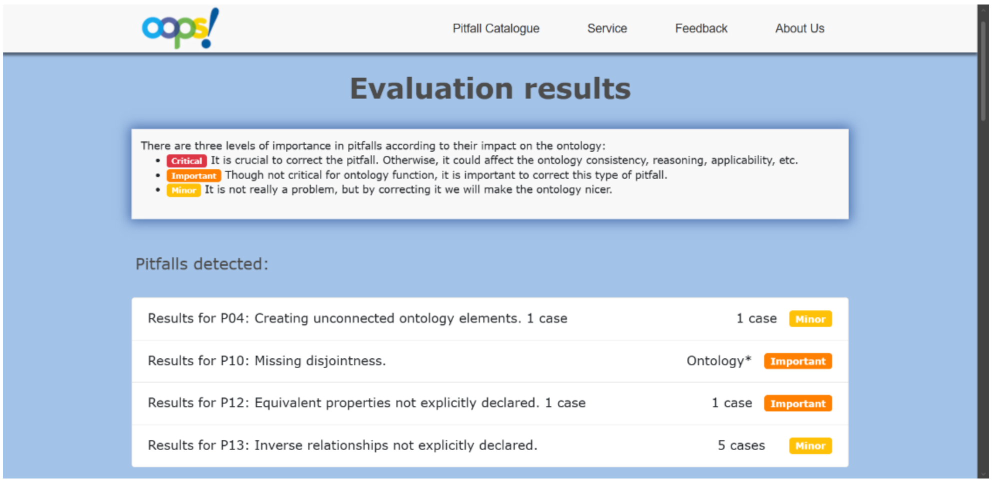
python -m pyshacl -s ./shacl/shapes.ttl -d ./data/tony_awards_data.ttl -m -i rdfs **Final validation result:**

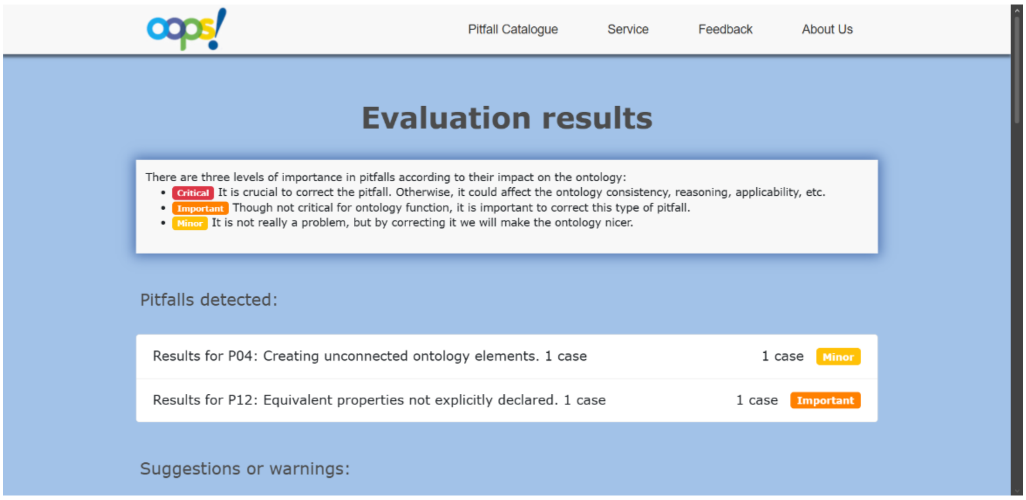
Conforms: True.



10. OOPS! evaluation

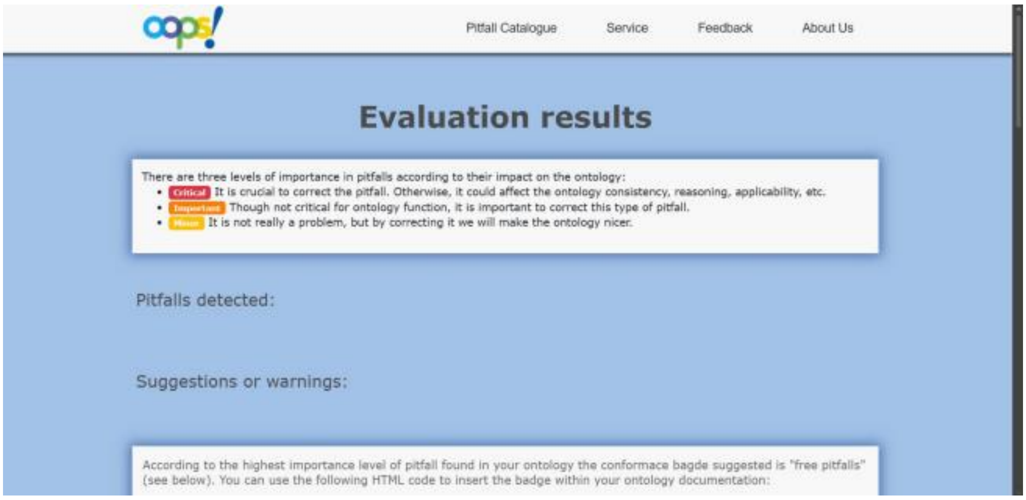
The ontology was evaluated with OOPS! to detect common ontology pitfalls. Initial evaluations detected issues such as disconnected ontology elements, missing disjointness, equivalent properties not explicitly declared and missing inverse relationships.





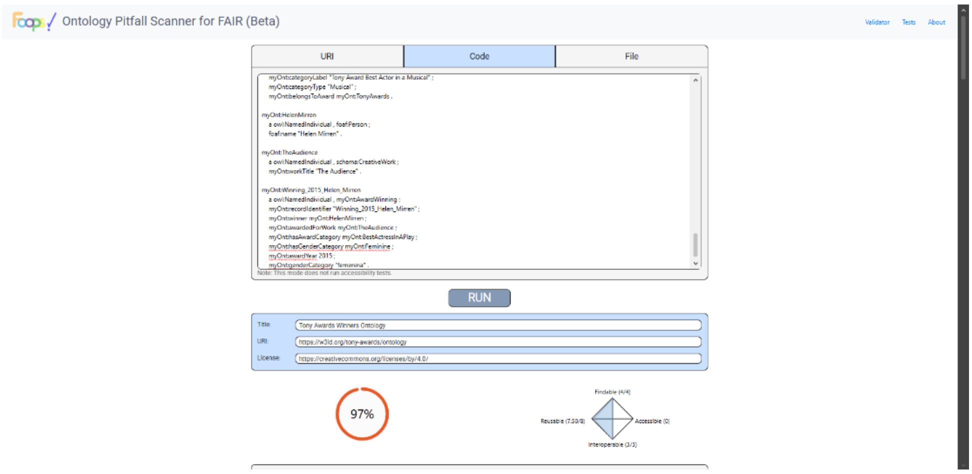
The ontology was improved by adding inverse properties, adding disjointness axioms, replacing duplicated name usage for creative works with myOnt:workTitle and improving annotations and metadata.

The final OOPS! evaluation reports no detected pitfalls.



11. FOOPS! evaluation and FAIRness

The ontology was evaluated with FOOPS! to assess its FAIRness. The final FOOPS! screenshot available in the repository reports a score of 97%. This evaluation was performed using ontology code/file input rather than a fully deployed resolvable URI.



12. Documentation and publication status

The repository includes ontology files, README documentation, SHACL documentation, evaluation screenshots and this final report. The ontology contains labels, comments and metadata to support documentation and reuse.

The ontology defines a w3id-based persistent URI: https://w3id.org/tony-awards/ontology. A pull request has been submitted to the official w3id.org repository to enable redirection and content negotiation for the ontology and resource namespaces: https://github.com/perma-id/w3id.org/pull/6089. At the time of delivery, the pull request is still pending review/merge, so the URI may still return Not Found until it is accepted.

13. Mapping with Group 9 ontology and SHACL adaptation

According to the assigned group pairing, Group 10 was paired with Group 9. Group 9 works on an Oscar/movie gender-analysis ontology, while this project models Tony Awards winners. The domains are different, but they share common concepts such as people, creative works, award records, award categories and gender categories.

Group 10 term	Group 9 term	Mapping decision
foaf:Person	oscar:Person	Equivalent at conceptual level: both

		represent people involved in award-related data
schema:CreativeWork	oscar:Movie	Close match: both represent a work associated with an award record, although Group 9 is specific to films.
myOnt:AwardWinning	oscar:AwardNomination	Close match: both act as event-like/n-ary records connecting award information.
myOnt:TonyAward	oscar:Award	Equivalent at a generic level, but referring to different award organizations.
myOnt:AwardCategory	oscar:Category	Equivalent at conceptual level.
myOnt:GenderCategory	oscar:Gender	Equivalent at conceptual level, but values need lexical normalization

To adapt the Group 10 SHACL shapes to the Group 9 dataset, a direct namespace replacement would not be enough because the two ontologies use different modelling structures.

The main target class should be changed from myOnt:AwardWinning to the Group 9 award record class, such as oscar:AwardNomination. The winner relation myOnt:winner should be replaced by the property used in Group 9 to link nominations to people, for example oscar:nominates if that property is available in their model.

The property myOnt:awardedForWork should be adapted to the movie relation used by Group 9, such as oscar:Movie or a property path through roles. The category relation myOnt:hasAwardCategory should be replaced by the corresponding Group 9 award or category property.

Gender validation also needs to be modified. In Group 10, gender is linked to the award-winning record through myOnt:hasGenderCategory and the literal property myOnt:genderCategory. In Group 9, gender is attached to people through oscar:hasGender, so the SHACL constraint should target oscar:Person instead.

Finally, myOnt:workTitle and myOnt:awardYear should be replaced by the corresponding movie title and ceremony/year properties available in the Group 9 ontology.

14. Adicional tools used

Several tools were used during the development of the project. Protégé was used to review and edit the OWL ontology. Chowlk was used to create the conceptual model diagram. pySHACL was used to validate the Knowledge Graph against the SHACL shapes. OOPS! was used to detect ontology modelling pitfalls, and FOOPS! was used to evaluate the FAIRness level of the ontology. GitHub was used to store and organize the project files. Finally, w3id.org was used to request persistent identifiers through a pull request, although the request was still pending review at the time of delivery.

15. Conclusions

The project produced a semantic resource for representing Tony Award winning records. The ontology includes more than 20 ontology elements, with 14 classes, 10 object properties and 7 datatype properties. The main design decision was the use of the myOnt:AwardWinning class as an n-ary relation connecting winner, work, category, gender category and year.

The ontology reuses FOAF, Schema.org, SKOS, Dublin Core Terms and VANN. SHACL validation was performed and the final report conforms. OOPS! reports no detected pitfalls, and FOOPS! reports a 97% score when evaluated from the ontology code/file.

The main publication limitation is that the w3id redirection and content negotiation are still pending review through the submitted pull request. In addition, the current repository RDF data file is a small conformant sample; a complete RDF materialization of all complete rows from the combined CSV would strengthen the Knowledge Graph deliverable.

16. Deliverables

Deliverable	File / URI
Ontology in Turtle	ontology/Ontology.ttl
Ontology in RDF/XML	ontology/Ontology.owl
SHACL shapes	shacl/shapes.ttl
Knowledge Graph	data/tony_awards_data.ttl
Functional requirements	requirements/Requirement.csv
Source dataset	data/dataset_combinado_.csv
Conceptual model	chowlk/diagrama_corregido.png
SHACL validation report	shacl/report.ttl
SHACL evidence	shacl/shacl_validation_conforms_true.png
OOPS! evidence	shacl/evaluation/oops/
FOOPS! evidence	shacl/evaluation/foops/
Ontology URI	https://w3id.org/tony-awards/ontology
Repository	https://github.com/tony07122004/group10-semantic-requirements